

Keyword Reference

	Keyword	Arguments	Description
Structure	deck		Begin a deck; end with "edeck"
	def	name args...	Define a function; end with "edef"
	for	x=begin end [increment]	Begin loop; end with "efor"
	if	condition...[else]...eif	Conditional; one of: a==b, a!=b, a>b, a<b, a>=b, a<=b, a<>b c
	import	"file"	import function found in a file
	include	"file"	Include the contents of a file
	slide	[bgcolor] [fgcolor]	Begin a slide; end with "eslide"
	edeck		End the deck
	edef		End the definition
	efor		End the for loop
	eif		End the conditional
	else		Begin the else clause
Utility	canvas	width height	Define with dimensions of the canvas
	content	"scheme://file"	Embed content
	dump	[name]	Dump variables
	grid	"file" x y hspace vspace edge	Define a content grid
	ruler	[increment] [color]	draw a (x,y) ruler
Graphics	acircle	x y w [color] [opacity]	Circle with size based on area
	arc	x y w h a1 a2 [lw] [color] [opacity]	Ellipical arc centered at (x,y), dimensions (w,h) between angles a1 and a2
	circle	x y w [color] [opacity]	Circle centered at (x,y), diameter w
	curve	bx by cx cy ex ey [lw] [color] [opacity]	Quadratic Bezier Curve begin (bx,by), control (cx, cy), end (ex,ey)
	ellipse	x y w h [color] [opacity]	Ellipse centered at (x,y), dimension (w,h)
	hline	x y w [lw] [color] [opacity]	Horizontal line begin at (x,y), length w
	line	x1 y1 x2 y2 [lw] [color] [opacity]	Line between (x1,y1) and (x2,y2)
	dline	x1 y1 x2 y2 [lw] [gap] [color] [opacity]	Dotted line between (x1,y1) and (x2,y2), size lw
	pill	x y w h [color]	Pill shape beginning at (x,y), dimensions (w,h)
	polygon	"x1 x2 x3...." "y1 y2 y3..." [color] [opacity]	Polygon with specified x, y coordinates
	polyline	"x1 x2 x3...." "y1 y2 y3..." [lw] [color] [opacity]	Polyline with specified x, y coordinates
	rect	x y w h [color] [opacity]	Rectangle centered at (x,y), dimensions (w,h)
	rrect	x y w h r [color] [opacity]	Rounded rectangle centered at (x,y), dimensions (w,h), corner radius r
	square	x y w [color] [opacity]	Square centered at (x,y), size w
	star	x y sides inner outer [color] [opacity]	Star centered at (x,y), with sides, innner and outer sizes
	vline	x y h [lw] [color] [opacity]	Vertical line beginning at (x,y), h high

	Keyword	Arguments	Description
Text	arctext	"string" x y radius a1 a2 fontsize [font] [color] [opacity] [link]	Text on an arc, at fontsize, center (x,y), radius r, between a1. a2
	btext	"string" x y fontsize [font] [color] [opacity] [link]	Text beginning at (x,y), at fontsize
	ctext	"string" x y fontsize [font] [color] [opacity] [link]	Centered text beginning at (x,y), at fontsize
	etext	"string" x y fontsize [font] [color] [opacity] [link]	End-aligned text at (x,y), at fontsize
	rtext	"string" x y angle fontsize [font] [color] [opacity] [link]	Rotated text centered at x,y), at angle and fontsize
	text	"string" x y fontsize [font] [color] [opacity] [link]	Text beginning at (x,y), at fontsize
	para	x y w fontsize [font] [color] [opacity] [link]...epara	Paragraphs beginning at (x,y), at fontsize with width w
	textblock	"string" x y w fontsize [font] [color] [opacity] [link]	Block of text beginning at (x,y), at fontsize, with width w
	textblockfile	"file" x y w fontsize [font] [color] [opacity] [link]	Block of text read for a file, beginning at (x,y), at fontsize, with width w
Lists	textcode	"file" x y w fontsize [font] [color] [opacity]	Lines of code, read from a file, upper right corner at (x,y), margin at w
	textfile	"file" x y fontsize [font] [color] [opacity] [spacing]	Contents of a text file upper right corner at (x,y)
	blist	x y fontsize [font] [color] [opacity] [spacing]	Bulleted list starting at (x,y), at fontsize
	clist	x y fontsize [font] [color] [opacity] [spacing]	Centered list starting at (x,y), at fontsize
	list	x y fontsize [font] [color] [opacity] [spacing]	List starting at (x,y), at fontsize
	nlist	x y fontsize [font] [color] [opacity] [spacing]	Numbered list starting at (x,y), at fontsize
Images	li	"item" [font] [color] [opacity]	List item
	elist		End the list
Images	cimage	"file" "caption" x y w h [scale] [link] capsize	Captioned image; center (x,y), dimensions (w,h) (h=0, w is % of canvas width)
	image	"file" x y w h [scale] [link]	Image center at (x,y), dimensions (w,h) (h=0, w is % of canvas width)
Braces/ Brackets	dbrace	x y w bw bh [lw] [color] [opacity]	Downward pointing brace
	dbracket	x y w h [lw] [color] [opacity]	Downward pointing bracket
	lbrace	x y h bw bh [lw] [color] [opacity]	Left pointing brace
	lbracket	x y w h [lw] [color] [opacity]	Left pointing bracket
	rbrace	x y h bw bh [lw] [color] [opacity]	Right pointing brace
	rbracket	x y w h [lw] [color] [opacity]	Right pointing bracket
	ubrace	x y w bw bh [lw] [color] [opacity]	Upward facing brace
	ubracket	x y w h [lw] [color] [opacity]	Upward facing bracket

	Keyword	Arguments	Description
Arrows	arrow	x1 y1 x2 y2 [lw] [aw] [ah] [color] [opacity]	Arrow starting at (x1,y1), ending at (x2,y2), aw=width, ah=height
	dcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Downward curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	lcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Left curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	rcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Right curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	ucarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Upward curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
Built-ins	x=area	expression	Assign an area
	x=format	"fmt" expr... (up to 5)	Assign formatting to expressions
	x=polar	x y radius angle	Assign polar coordinate centered at (x,y) at radius and angle (0-360)
	x=polarx	x y radius angle	Assign X-polar coordinate centered at (x,y) at radius and angle (0-360)
	x=polary	x y radius angle	Assign Y-polar coordinate centered at (x,y) at radius and angle (0-360)
	x=random	min max	Assign a random number between two values
	x=substr	"string" begin end	Assign a substring
	x=vmap	data min1 max1 min2 max2	Assign a value mapped to two ranges
Math	x=geocoord	geo URL or lat long	Map geo to canvas coordinates
	x=cosine	expression	Assign the cosine of expression
	x=sine	expression	Assign the sine of expression
	x=sqrt	expression	Assign the square root of expression
	x=tangent	expression	Assign the tangent of expression
Geographic	geoarc	"p1" "p2" [lw] [color] [opacity]	Draw arcs between points
	geoborder	"file" [lw] [color] [opacity]	Reads KML data from the specified file and renders the map borders
	geoimage	"loc" width height	Place an image at a geographical location
	geolabel	"loc" [size] [font] [color] [opacity]	Reads data from the specified file or location and renders the map labels
	geoloc	"loc" [align] [size] [font] [color] [opacity]	Reads data from the specified file or location and a make map point and labels
	geomark	"loc" [size] [color] [opacity]	Reads data from the specified file or location and renders map points
	geopath	"p1" "p2" [lw] [color] [opacity]	Draw line between points
	geopathfile	"file" [lw] [color] [opacity]	Reads data from the specified file and a make lines between points
	georegion	"file" [color] [opacity] [gap]	Reads KML data from the specified file and renders the map regions
	geobound	latmin [latmax] [longmin] [longmax]	Sets overall geographic bounds
	geocanvas	xmin [xmax] [ymin] [ymax]	Sets geographic canvas

	Keyword	Arguments	Description
Charts	dchart	options... file	Chart with specified options
	legend	"string" x y fontsize font color	Chart legend
	chartbbox	left [right] [top] [bottom]	Chart bounding box
	areachart	"file" [color] [lcolor] [vcolor]	Area chart with specified data
	barchart	"file" [color] [lcolor] [vcolor]	Bar chart with specified data
	dotchart	"file" [color] [lcolor] [vcolor]	Dot chart with specified data
	hbarchart	"file" [color] [lcolor] [vcolor]	Horizontal bar chart with specified data
	linechart	"file" [color] [lcolor] [vcolor]	Line chart with specified data
	wbarchart	"file" [color] [lcolor] [vcolor]	Word bar chart with specified data
	slopechart	"file" [color] [lcolor] [vcolor]	Slope chart with specified data
	pmap	"file" [size]	Proportional map chart with specified data
	pie	"file" [size]	Pie chart with specified data
	donut	"file" [width] [size]	Donut chart with specified data
	pgrid	"file" [size]	Proportional grid chart with specified data
	waffle	"file" [size]	Waffle chart with specified data
	fanchart	"file" [size]	Fan chart with specified data
	bowtie	"file" [size]	Bowtie chart with specified data